

ITAI-1371 - Intro to Machine Learning
Reflective Journal

Camila Ferreira da Silva

In Lab 04, I explored the purpose of Exploratory Data Analysis (EDA) through the Titanic dataset. My main contribution was combining the group members' sections into one notebook and reviewing the complete assignment. This gave me the opportunity to look at the lab as a whole and consider how the statistics, visualizations, and explanations connected. It helped me understand that data analysis involves more than running code: we also need to interpret the results and communicate what they mean.

One of the main lessons I took from this lab was the importance of understanding a dataset before building a machine learning model. Examining the first rows, column types, and missing values provides essential information about the data. For example, missing ages can affect how we interpret age distributions because those graphs do not represent every passenger. This reminded me that conclusions depend on the information available and that missing data should be considered rather than overlooked.

The lab also helped me understand why summary statistics and visualizations should be used together. A mean or a count provides a useful numerical overview, but it cannot show every detail of a distribution. Graphs make it easier to notice differences between groups, unusual values, and patterns that may deserve further investigation. Comparing survival across passenger classes and sexes illustrates how looking at groups separately can reveal information that an overall survival percentage does not explain.

Another important point was the difference between counts and rates. A group may have more survivors simply because it contains more passengers. To compare survival chances fairly, we need to consider the proportion of survivors within each group. This is especially relevant when examining passenger class or port of embarkation. I also learned to be careful with explanations: a relationship between a variable and survival does not automatically mean that the variable caused the outcome. Other factors may help explain the pattern.

Combining and reviewing the group's work made me think more carefully about the purpose of each section. I needed to consider whether the explanations matched the questions and whether the notebook communicated a consistent analysis. Reviewing the whole assignment encouraged me to move beyond individual answers and follow the reasoning from the initial exploration to the final observations. It also reinforced the importance of allowing time for a final review when working in a group.

Overall, this lab strengthened my understanding of EDA as a foundation for machine learning. I learned to pay closer attention to missing values, group sizes, distributions, and the limits of an interpretation. In future assignments, I want to focus on explaining why I choose a particular visualization and supporting my conclusions with appropriate evidence from the data.

Great Inioluwa Olayiwola

When I first started looking at the end-to-end Machine Learning (ML) workflow, it honestly felt like a lot to take in. You're jumping between loading CSV files, calculating stats, filling in missing data, building Seaborn charts, and setting up training pipelines. At first, it just seemed like an intimidating checklist of steps to get through. But once I started working through the Titanic dataset, things really clicked: I stopped seeing ML as a rigid list of tasks and started seeing it as a logical, step-by-step story where every phase directly shapes the next.

I realized that Exploratory Data Analysis (EDA) isn't just something you do to make pretty charts before the real work begins. It's the foundation of the whole process. You can't build a reliable predictive model without first understanding the layout, weird quirks, and hidden connections in your data. Running initial checks with `df.info()` and `df.describe()` brought some huge details to light. For instance, seeing that 177 values were missing in Age and that Cabin was mostly empty (only 204 non-null entries out of 891) completely changed how I thought about feature engineering. Instead of guessing how to fix missing data, I could plan a targeted strategy. Without taking the time to diagnose these issues upfront, feeding messy data straight into a model would just lead to bad predictions, classic "garbage in, garbage out."

My biggest lightbulb moment came when I compared basic summary statistics to visual charts. At first, I thought simple numbers like the mean, standard deviation, and quartiles were enough to understand a dataset. But analysing the Titanic passenger data proved just how much single summary numbers can hide.

Take passenger age, for example. The overall average age came out to 29.7 years. If you just looked at that number, you'd probably picture a standard bell curve centered around young adults. But when I built side-by-side histograms using FacetGrid to compare survivors and non-survivors, a huge detail popped out. There was a massive spike in survival for infants and young kids under 10. That critical detail completely disappears if you only look at the average age.

We saw the exact same thing with passenger fares. The mean fare was \$32.20, but the raw numbers showed a minimum of \$0.00 and an extreme maximum of \$512.33. Creating a boxplot of Fare versus Survived made the heavy skew obvious right away. The median fare for people who died was around \$10.00, while the distribution for survivors stretched way higher, with an interquartile range going past \$50.00 and plenty of high-value outliers. A single average fare of \$32.20 gave a totally misleading picture for both groups. Visualizing the data made everything clear and showed how much socioeconomic status influenced who got access to lifeboats.

Digging into categorical features gave me a much better grasp of how feature interactions drive classification models. Plotting the target variable with `sns.countplot()` laid out the baseline reality: 549 passengers died while 342 survived, giving a baseline survival rate of 38.4%. Spotting this class imbalance early is key for model design. If you built a simple model that just guessed "Died" for every single person, it would claim an accuracy of 61.6%, even though it completely failed at predicting anything useful.

Breaking survival down by different variables brought out some really strong patterns:

1. Gender (Sex): Over 70% of female passengers survived, compared to under 20% of male passengers, making gender the strongest individual predictor in the dataset.
2. Passenger Class (Pclass): More than 60% of 1st Class passengers survived, while 3rd Class passengers saw a survival rate under 25%. This clearly showed how ticket class played a role during the evacuation.
3. Port of Embarkation (Embarked): Passengers who boarded at Cherbourg (C) had a noticeably higher survival rate than those who got on at Southampton (S) or Queenstown (Q). But looking closer, this wasn't really about geography, it turned out Cherbourg simply had a much higher percentage of 1st Class passengers.

Karima Sadruddin

Before this lab, I thought data analysis mostly meant looking at numbers and calculating averages. This lab helped me understand that exploratory data analysis (EDA) is an important step before building a machine-learning model. EDA helps us learn what the data contains, notice missing values, and look for patterns that might affect a model's results.

The Titanic dataset made the ideas easier to understand because it included passenger information such as age, gender, passenger class, fare, and survival. The notebook uses summary statistics and several plots to compare these details. A count plot can show how many passengers survived or died. Comparing survival by class and gender helps reveal differences between groups. A histogram helps us look at age distributions, while a box plot can compare ticket fares for passengers who survived and those who did not.

One important thing I learned is that a chart and a summary number answer different questions. An average age gives one useful number, but it does not show the full distribution or whether different groups have different patterns. A plot can make differences, outliers, and skewed data easier to notice. Using both statistics and visualizations gives a better understanding than relying on only one method.

The experimentation section asks us to compare survival by port of embarkation and compare ticket fares by survival outcome. These questions helped me practice choosing a visualization based on the type of data. A count plot is useful for categories such as embarkation port, while a box plot is useful for comparing a numerical variable such as fare across two outcome groups. I also included survival proportions by port because the number of passengers from each port is not the same. Proportions can make group comparisons fairer than raw counts alone.

This lab also reminded me that a relationship does not prove cause and effect. If passengers with higher fares or certain passenger classes had better survival outcomes, fare itself may not be the only reason. Fare can be connected to passenger class and other circumstances. In machine learning, it is important to question what the data actually supports instead of making a conclusion too quickly.

The workflow makes more sense to me now: first load the data, then inspect its structure and missing values, summarize it, visualize important variables, and use the findings to guide later modeling decisions. EDA does not guarantee that a model will be accurate, but it can help identify data quality issues and useful features before modeling begins. My next step is to run all notebook cells in Google Colab, check the actual outputs, and make sure my written interpretations match the charts and values produced.

Quan Thach

Completing this lab made me feel like I was given an investigation file and told to deduce the story. At the beginning of this notebook, it says that EDA is "like a detective's job," and this thought lingered in my mind throughout the exercise. The dataset used in this lab was the Titanic passenger list, which had 891 rows and columns like Survived, Class, Sex, Age, Fare, and Embarked. I previously believed that data analysis was merely a preliminary step in training the model. This notebook showed me that the first task of a data analyst is to look at the data, ask questions, and wait for the data to answer.

What I first noticed is how the setup cell provided some important information even before I started plotting. Thanks to the `df.head()` output, I could see what the single passenger looks like as well as the `.describe()` output in Part 2 that provided data like an average of 29.7 as a number age and an overall survival rate of 38.4%. I also noticed that the Age count was 714 while the other columns showed 891, which indicated to me that some data was missing. I also noticed that Age and Cabin had missing values, which were clearly mentioned in the notebook and are a reminder that real data is never entirely clean. I encountered a little trouble in that cell with the `!df.info()` command. It caused an error because the `!` symbol tells the Jupyter notebook to use the shell instead of Python.

Part 3 was where the lab really came alive for me. The first countplot showed that far more people died than survived, and the notebook called this an "imbalanced dataset," which was a new concept for me. Then the class plot made the pattern hard to ignore: 1st class passengers survived at a much higher rate than 3rd class passengers. The gender plot was even more striking, with the notebook calling the pattern "undeniable" because a much higher proportion of females survived than males. Finally, the age histograms, built with FacetGrid, showed a spike of young children among the survivors, which supported the "women and children first" idea the lab mentioned. What I appreciated most was how each plot came with a one-line insight printed underneath, which trained me to stop and say what a chart actually means instead of just admiring it.

The experimentation section is where I had to think for myself. In Experiment 1, I looked at the port of embarkation. The countplot showed Southampton with by far the most passengers, but raw counts can mislead, so I also calculated survival rates: about 55.4% for Cherbourg, 39.0% for Queenstown, and 33.7% for Southampton. That was a good lesson in not trusting a bar chart on its own. It also made me suspect that the port was not the real cause, since the class pattern from earlier might be hiding behind it. In Experiment 2, I compared fares for survivors and non-survivors with a boxplot. The first version was squashed because a few tickets cost hundreds of pounds, so I added a log-scale version, and the difference became much clearer. The `.describe()` output backed it up: the median fare was 10.5 for those who

died and 26.0 for those who survived, while the maximum fare of 512.33 showed how far the outliers stretched.

The knowledge check at the conclusion puts all information together. In my response, I concluded that the main idea of EDA is to obtain knowledge regarding the data prior to its modelling, that the young girl or a woman travelling in 1st class has the best opportunity to win, and that visualization is essential as descriptive statistics do not reveal the shape, outliers, and differences among groups. The diagrams of charges serve as an excellent example for the last point. The average charge for people who survived was 48.4, while the median charge was only 26.0, which I would never understand without the examination of the graph. All in all, I think that the most important lesson for me was that there are seldom isolated patterns in data. Class, charge, and port of departure seemed to tell interconnected stories, and the laboratory instructed me to check if one variable causes another or just indicates its existence. After this laboratory, I feel more self-assured when reading graphs, more vigilant regarding the difference between raw counts and rates, and more interested in what else lies across the unexplained columns.